

Efficient Media Heating through Optical Energy-Flow Control via NFT and Spacing Design in HAMR

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Heat-assisted magnetic recording (HAMR) requires efficient and localized energy delivery to the recording medium. Electromagnetic energy transport in a HAMR head system is analyzed using ADE-FDTD numerical simulations combined with an energy-flow analysis based on Poynting's theorem. The analysis focuses on the optical properties of the NFT and the spacing layer between the NFT and the recording medium. The results show that introducing a localized dielectric spacing layer significantly enhances heat generation in the recording medium. This enhancement originates from the formation of a resonant energy-coupling mode among the NFT peg, the spacing layer, and the recording layer. These results indicate that optical design, especially of the spacing layer, is important for controlling energy delivery to the recording medium in HAMR.

Index Terms— hard disk drive (HDD), heat-assisted magnetic recording (HAMR), near-field transducer (NFT), surface plasmon resonance (SPR), electromagnetic energy flow

I. INTRODUCTION

Heat-Assisted magnetic recording (HAMR) is a key technology for extending magnetic recording beyond the limits imposed by thermal stability and writability. By locally heating the recording medium during the write process, HAMR enables reliable magnetization reversal while maintaining long-term thermal stability in high-anisotropy media.

In HAMR, optical energy from a laser integrated into the recording head is delivered to the recording medium through a near-field transducer (NFT), which concentrates electromagnetic energy into a nanoscale region. To achieve higher recording density, it is crucial to clarify how electromagnetic energy is transported to the recording medium through the NFT-media system.

In this study, the energy flow, heat generation, and resonant coupling behavior in the spacing layer are analyzed using a time-domain electromagnetic energy-flow analysis based on Poynting's theorem. The results show that the optical properties of the spacing layer between the NFT and the recording medium play an important role in energy transfer, in addition to the geometric design of the NFT.

II. MODEL AND METHOD

The auxiliary differential equation finite-difference time-domain (ADE-FDTD) method^[1] was used for electromagnetic simulations to investigate energy transfer in a HAMR head. As shown in Fig. 1, the simulation model consists of a waveguide, a lollipop-type NFT with a peg, a main pole (MP), a spacing layer (SL), and a FePt recording layer (RL).

The incident light was a transverse magnetic (TM) polarized electromagnetic wave with a wavelength of 800 nm. Ta₂O₅ ($\epsilon = 4.5$) and Al₂O₃ ($\epsilon = 2.72$) were used as the core and cladding materials of the waveguide, respectively. The NFT was assumed to be either Au or Al. The complex dielectric constants of Au and Al are $\epsilon = -20.3 + 2.07i$ and $\epsilon = -62.4 + 44.5i$ at 800 nm, respectively. Au exhibits plasmonic behavior, whereas Al does not and was therefore used as a non-resonant reference

material. The MP was FeCo ($\epsilon = -77.6 + 37.6i$), and the RL was FePt ($\epsilon = -67.2 + 25.9i$). A Au heat-sink layer was placed beneath the recording layer.

The permittivity of the SL, ϵ_d was changed for 1, 3, and 5 to investigate the role of the spacing layer in energy transfer to the recording layer. Two types of spacing-layer configurations were investigated: a uniform layer over the entire recording-medium interface (Fig. 1(b)) and a local dielectric layer just beneath the NFT peg with an air ($\epsilon_d = 1$) matrix (Fig. 1(c)).

To analyze energy flow in the system, Poynting's theorem was employed, which describes the conservation of electromagnetic energy^[2]. It is expressed as

$$\nabla \cdot (\mathbf{E} \times \mathbf{H}) = - \left[\frac{\partial W_e}{\partial t} + \frac{\partial W_h}{\partial t} + P \right].$$

Here, the electric energy density W_e , magnetic energy density W_h , and heat generation P are defined as

$$W_e = \frac{\epsilon_0 \epsilon_\infty |\mathbf{E}|^2}{2}, W_h = \frac{\mu_0 |\mathbf{H}|^2}{2} + \frac{|\mathbf{J}_p|^2}{2\epsilon_0 \omega_p^2}, P = \frac{v_c |\mathbf{J}_p|^2}{\omega_p^2 \epsilon_0}.$$

ϵ_0 and μ_0 are permittivity and permeability of vacuum, ϵ_∞ is the high-frequency dielectric constant, $\mathbf{E}$ and $\mathbf{H}$ are the electric and magnetic fields, $\mathbf{J}_p$ is the polarization current, ω_p is the plasma frequency, and v_c is the collision frequency.

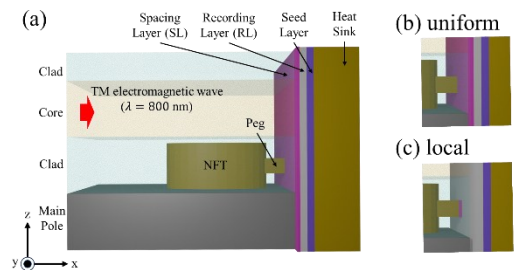


Fig. 1. (a) Simulation model used in this study. (b) Uniform dielectric layer configuration, where a dielectric layer with permittivity ϵ_d is placed over the entire recording-medium interface. (c) Local spacing configuration, where the dielectric (ϵ_d) is placed only beneath the NFT peg, while the surrounding region is air ($\epsilon_d = 1$).

III. RESULT AND DISCUSSION

Figure 2 shows the heat generation in the recording medium for Au- and Al-NFTs. Figures 2(a) and 2(c) correspond to the Au-NFT, while Figs. 2(b) and 2(d) correspond to the Al-NFT. Figures 2(a) and 2(b) show the total heat generation integrated over a $100 \text{ nm} \times 100 \text{ nm}$ area beneath the peg and the full thickness of the RL, whereas Figs. 2(c) and 2(d) show the surface distribution along the down-track direction.

For the case of the uniform SL with $\epsilon_d = 1$, the Au-NFT produces larger heat generation in the medium compared to the Al-NFT. When $\epsilon_d = 3$, the heat generation decreases for the Au-NFT, while it remains almost unchanged for the Al-NFT. In contrast, for the local dielectric configuration of the SL, a pronounced enhancement of heat generation is observed only for the Al-NFT. As shown in Figs. 2(c) and 2(d), heat generation is localized just below the peg area in the local configuration. These results indicate that a localized dielectric structure in the SL is key to generating a small heat spot in the recording medium.

The time-domain behavior of energy flow was investigated to clarify the relationship between heat generation and electromagnetic energy transport. The energy flow, defined as $EF = \partial W_e / \partial t + \partial W_h / \partial t$, is evaluated at the peg tip and within the spacing layer. Figures 3(a) and 3(b) show the time evolution of the EF for the Au-NFT and Al-NFT, respectively, under the two spacing-layer configurations.

For the case of Au-NFT under the uniform configuration, EF exhibits a clear anti-phase oscillation between the peg tip and the spacing layer. This behavior indicates strong energy coupling among the peg, the SL, and the RL and confirms the formation of a resonant coupling mode, as illustrated in Fig. 3(c). In contrast, for the Al-NFT, no resonant coupling is observed under the uniform spacing configuration, as shown in Fig. 3(d). However, a clear energy coupling emerges in the local

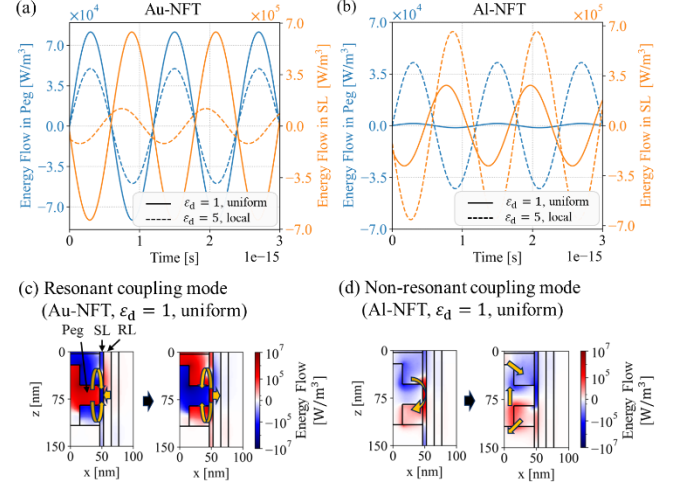


Fig. 3. (a) Time-domain energy-flow term evaluated at the NFT peg tip and within the spacing layer for the Au-NFT, and (b) for the Al-NFT. The solid lines represent the uniform spacing configuration with $\epsilon_d = 1$, while the dashed lines represent the localized spacing configuration with $\epsilon_d = 5$. (c) Energy-flow distribution in the resonant coupling mode, (d) in the non-resonant coupling mode.

configuration, indicating the formation of a resonant coupling state.

Importantly, this resonant coupling behavior is consistent with the heat-generation results shown in Fig. 2, where enhanced heating in the recording medium is observed under the same conditions. These results indicate that strong energy coupling among the peg, the SL, and the RL plays a key role in enabling efficient electromagnetic energy transfer to the recording medium.

IV. CONCLUSION

Electromagnetic energy transport from a HAMR head to the recording medium was analyzed using time-domain simulations combined with an energy-flow analysis. The results show that heat generation in the medium is affected by the dielectric properties of the NFT peg and the spacing layer beneath it. They also indicate that the formation of a resonant coupling mode among the peg, the SL, and the RL is important for achieving efficient heating. These findings suggest that energy-flow optimization enables more flexible NFT design beyond conventional plasmonic materials.

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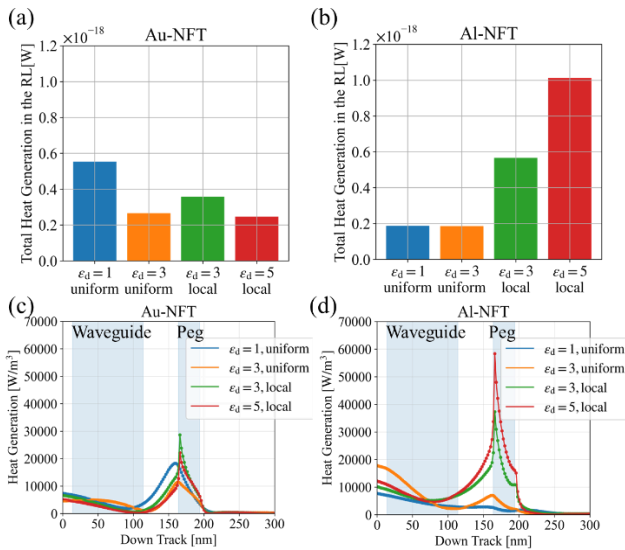


Fig. 2. (a) Total heat generation within a $100 \text{ nm} \times 100 \text{ nm}$ area beneath the NFT peg for the Au-NFT, comparing uniform and local spacing conditions with $\epsilon_d = 1, 3$ and 5 , (b) For the Al-NFT. (c) Down-track distribution of heat generation at the top surface of the recording layer (RL) for the Au-NFT under the same conditions as in (a), (d) for the Al-NFT.